

Frequency-free Dynamical Screening Bethe-Salpeter Equation

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SAUDI ARABIA'S JOURNEY TOWARD A CO₂-FREE, CARBON-FREE FUTURE

VISION 2030 LAYING THE FOUNDATION, VISION 2060 ACHIEVING A SUSTAINABLE LEGACY

Guided by the leadership of His Royal Highness Crown Prince Mohammed bin Salman,
Saudi Arabia is transforming today for a thriving, sustainable tomorrow.

"We will not
live in oil forever.
We will develop
our country to be
stronger and more
sustainable."

- MBS

"Our goal is a
vibrant society,
a thriving economy,
and an ambitious
nation in a
greener world."

- MBS



THE MIGRATORY PATH: FROM CARBON-INTENSIVE TO CO₂-FREE, CARBON-FREE FUTURE



MBS' VISION FOR SAUDI ARABIA

- A strong economy built on innovation and diversification
- A sustainable environment for generations to come
- An ambitious nation leading global transformation
- A society where everyone can thrive and contribute

KEY ENABLERS



A GREENER SAUDI, A BETTER WORLD

- Cleaner air and a healthier environment
- Sustainable use of natural resources
- Resilient economy for long-term prosperity
- Positive impact for the region and the world

ONE PLANET. ONE FUTURE. OUR RESPONSIBILITY.

OUR COMMITMENT
TO THE PLANET



NET ZERO
EMISSIONS
BY 2060



50% RENEWABLE
ENERGY CAPACITY
BY 2030



IMPROVE ENERGY
EFFICIENCY BY 30%
BY 2030



REDUCE EMISSIONS
INTENSITY BY 278%
BY 2030



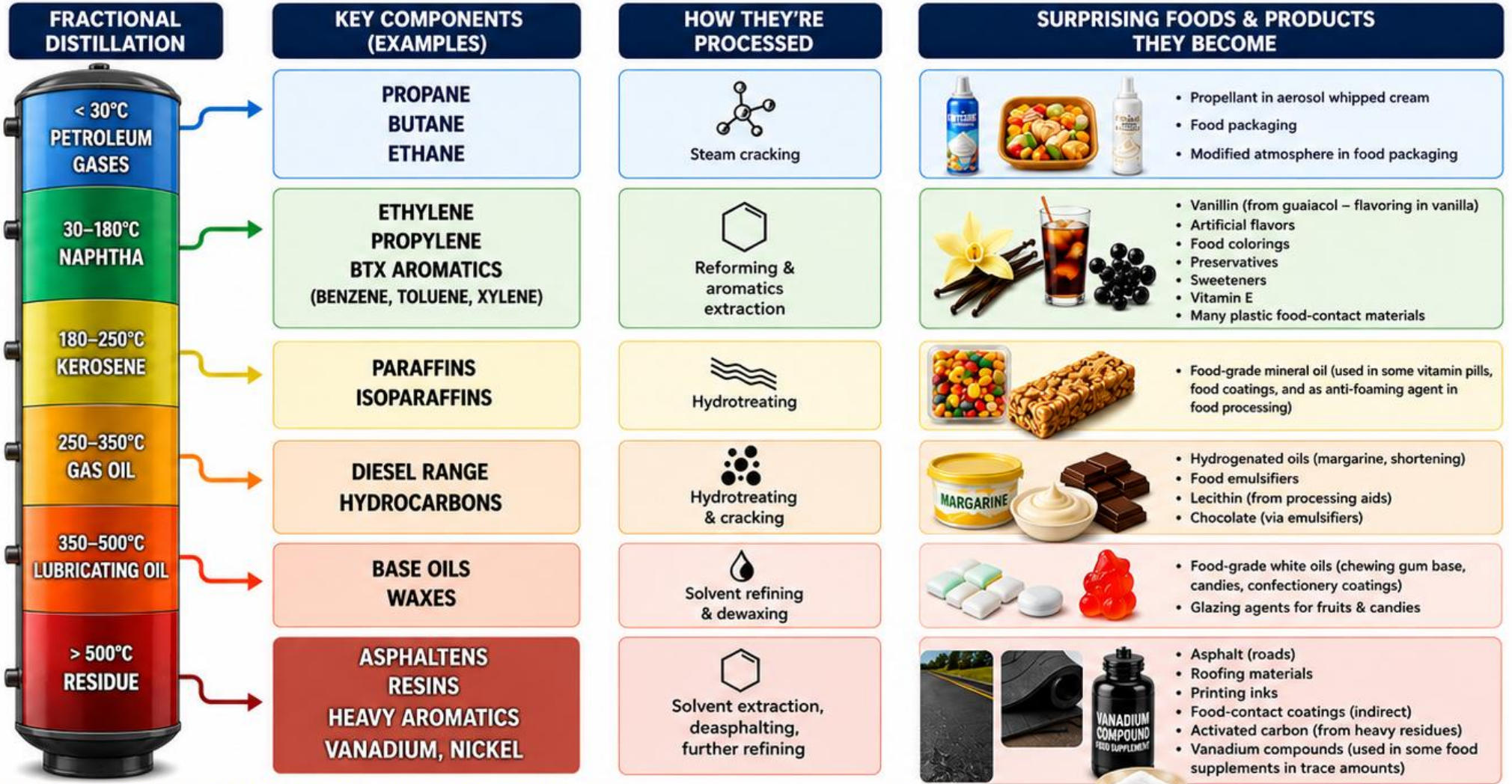
PLANT 10 BILLION
TREES ACROSS THE
KINGDOM

TOGETHER, WE BUILD A THRIVING,
SUSTAINABLE SAUDI ARABIA.



BEYOND FUEL: SURPRISING FOODS & EVERYDAY PRODUCTS FROM CRUDE OIL

Crude oil is refined into many components that become products we use and eat every day.



THE "RESIDUE" — THE PART PEOPLE LEAST EXPECT

What's left after refining (black, sticky, heavy) is still extremely valuable.

It contains:

- Asphaltenes
- Resins
- Heavy aromatics
- Metals (like vanadium, nickel)



EXAMPLES YOU MIGHT NOT EXPECT



Chewing gum base



Vanilla flavor (vanillin)



Candy glaze



Anti-foaming agents



Some vitamin supplements



Chocolate (emulsifiers)



Asphalt for infrastructure

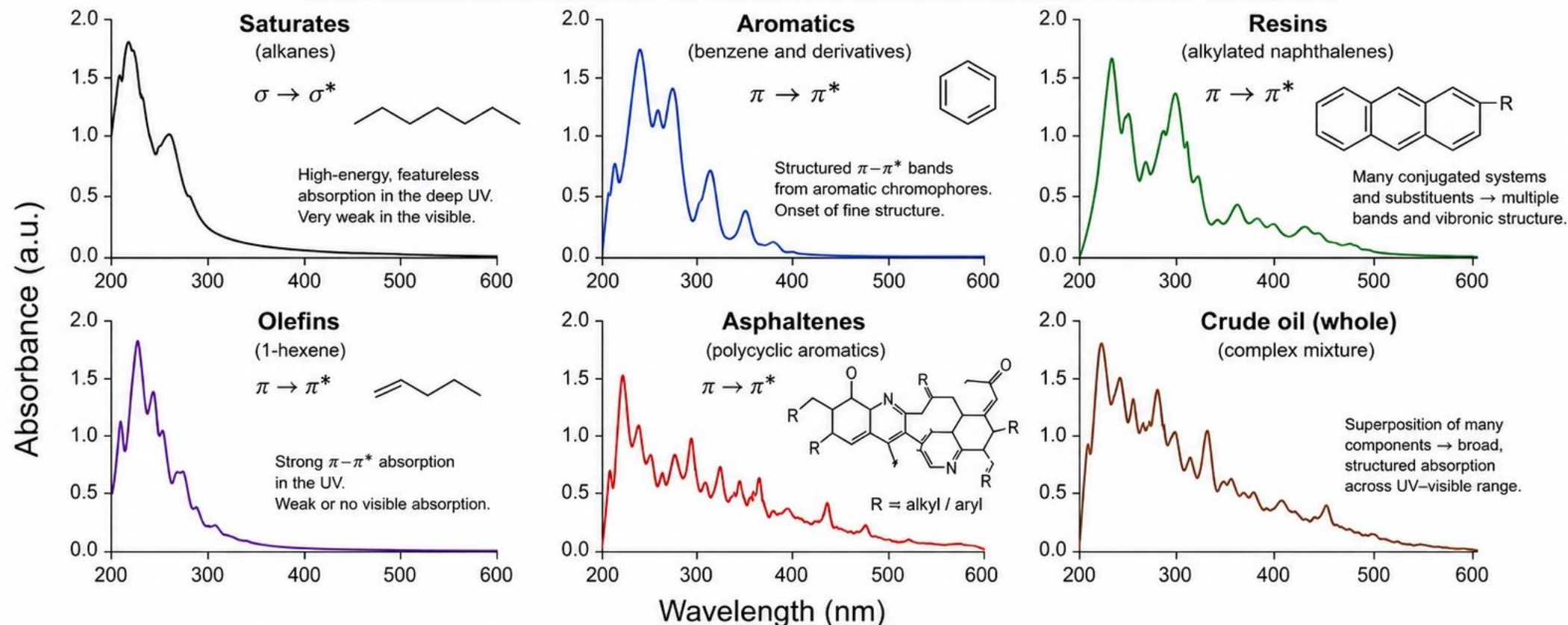


Crude oil touches far more of our food system than most people realize.

Optical absorption spectra of different components of crude oil

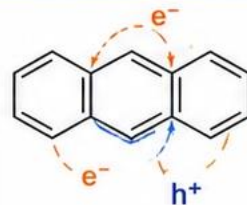
Complex spectra arise from overlapping molecular transitions and Frenkel-type excitations in aromatic and polycyclic organic fractions

These are schematic spectra (not quantitative experimental UV-Vis data) illustrating typical trends.



Physical origin of the complex spectra

Aromatic and polycyclic molecules in crude oil contain extended π -conjugated systems.



Photoexcitation creates electron-hole pairs that are strongly localized on molecular units (**Frenkel-type excitations**).

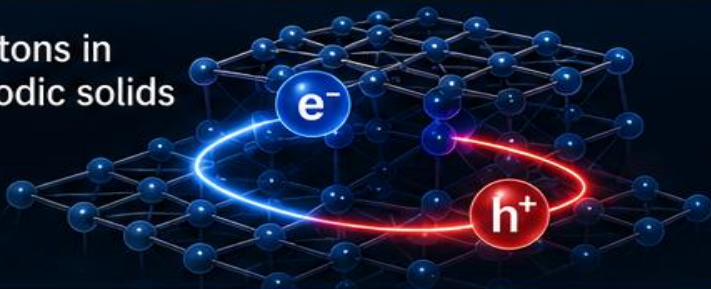
Coulomb interaction between neighboring chromophores leads to **Frenkel-type excitons** (coherent delocalization of the excitation).

Multiple allowed molecular and excitonic transitions with different energies, together with vibronic structure and chemical disorder, produce **broad structured absorption bands**.

Frenkel-type excitations are most relevant for aromatic, resin and asphaltene fractions of crude oil. Saturates show mainly high-energy $\sigma \rightarrow \sigma^*$ absorption and are not Frenkel-exciton dominated.

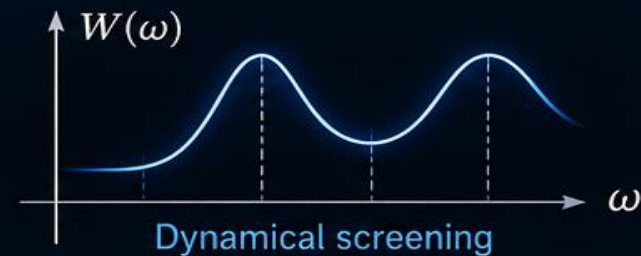
Frequency-free dynamical-screening Bethe–Salpeter equation

Excitons in
periodic solids



Dynamical BSE (nonlinear in Ω)

$$\left(H_{stat} + \underbrace{K^d(\Omega)}_{\text{dynamical kernel}} \right) A = \Omega A$$

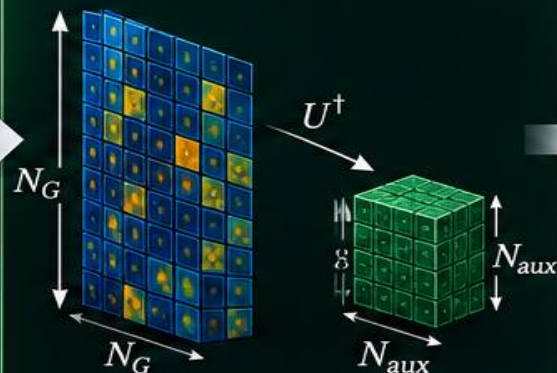


1 Microscopic screening
with local fields
 $W_{GG'}(q, \omega)$



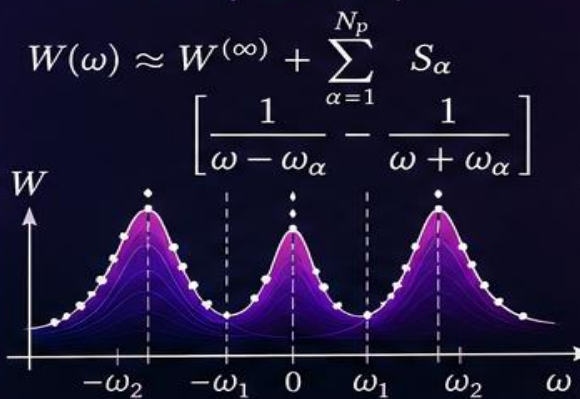
Full local-field (LF) matrix
 $N_G \times N_G$

2 Local-field compression
 $U^\dagger W^{sym} U$



Project to dielectric subspace
 $N_{aux} \ll N_G$

3 Causal multi-pole screening
in compressed space



Systematic & controllable in N_p
(positive-semidefinite residues S_α)

4 Frequency-free
generalized eigenproblem
 $\mathcal{H}_{aux} X = \Omega S X$



Single linear eigenproblem
→ bound exciton poles Ω_S



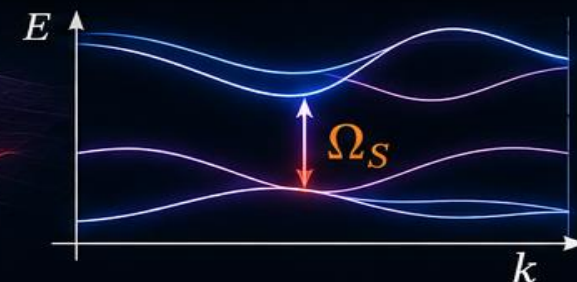
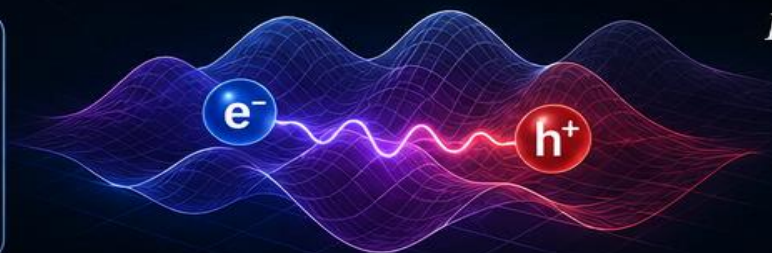
Full microscopic local fields



Controlled convergence in (N_{aux}, N_p)



No frequency scans: low-lying excitons from one eigenproblem



STEP 1

Symmetrized screening and local-field compression

Build the symmetrized screened interaction $W^{\text{sym}}(\mathbf{q}, \omega)$ and project the full local-field matrix onto a low-rank dielectric subspace

1. What we compute

For each momentum transfer $\mathbf{q}$ and frequency ω , we compute the microscopic screened interaction in the plane-wave (local-field) basis $\{\mathbf{G}\}$:

$$W_{GG'}(\mathbf{q}, \omega)$$

with $\mathbf{G}, \mathbf{G}'$ reciprocal lattice vectors.

2. Symmetrized screened interaction

Bare Coulomb is diagonal in $\mathbf{G}$,

$$V(\mathbf{q}) = \text{diag}\{v_G(\mathbf{q})\}, \quad v_G(\mathbf{q}) = \frac{4\pi}{|\mathbf{q} + \mathbf{G}|^2}$$

Inverse dielectric matrix $\epsilon_{GG'}^{-1}(\mathbf{q}, \omega)$ is dense.

Define the symmetrized screened interaction

$$W^{\text{sym}}(\mathbf{q}, \omega) = V^{1/2}(\mathbf{q}) \epsilon^{-1}(\mathbf{q}, \omega) V^{1/2}(\mathbf{q})$$

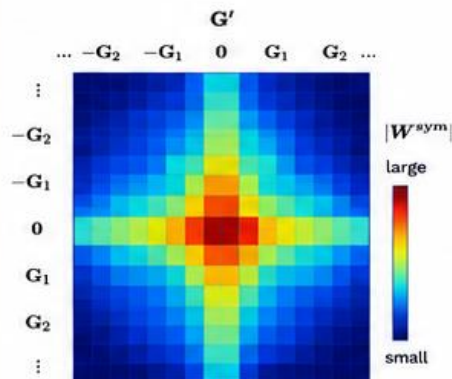
Elementwise,

$$W_{GG'}^{\text{sym}}(\mathbf{q}, \omega) = \sqrt{v_G(\mathbf{q})} \epsilon_{GG'}^{-1}(\mathbf{q}, \omega) \sqrt{v_{G'}(\mathbf{q})}$$

Since $W(\mathbf{q}, \omega) = \epsilon^{-1}(\mathbf{q}, \omega) V(\mathbf{q})$, we have $W^{\text{sym}} = V^{1/2} W V^{-1/2}$. Thus W^{sym} and W contain exactly the same physical screening information (similarity transformation). This balances the Coulomb weights on both indices, improving numerical conditioning and making the Hermitian/positivity structure more transparent.

3. Matrix structure in G-space

$W^{\text{sym}}(\mathbf{q}, \omega) \in \mathbb{C}^{N_G \times N_G}$ is generally dense because of local-field mixing.



- Off-diagonal elements encode coupling between different $\mathbf{G}$ and $\mathbf{G}'$ (local-field mixing)
- Strong central head element ($\mathbf{G} = \mathbf{G}' = 0$) controls long-range screening at small $\mathbf{q}$
- All Fourier components $\mathbf{q} + \mathbf{G}$ couple to $\mathbf{q} + \mathbf{G}'$ through screening

4. Local-field compression (dielectric subspace)

4a. Dielectric subspace

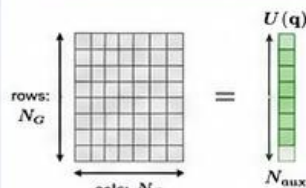
Introduce, for each $\mathbf{q}$, a low-rank subspace spanned by columns of $U(\mathbf{q})$:

$$U(\mathbf{q}) \in \mathbb{C}^{N_G \times N_{\text{aux}}}, \quad N_{\text{aux}} \ll N_G$$

$$U^\dagger(\mathbf{q}) U(\mathbf{q}) = I_{N_{\text{aux}}}$$

and the orthogonal projector

$$P(\mathbf{q}) = U(\mathbf{q}) U^\dagger(\mathbf{q})$$



- Columns of $U(\mathbf{q})$ span the dominant microscopic screening directions (local-field eigenpotentials)
- N_{aux} reduces the local-field dimension without any approximation to the frequency dependence
- Only the G-space sector is reduced

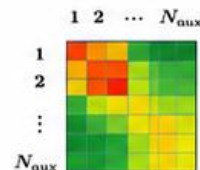
4b. Projection (compression)

Approximate W^{sym} by its action within the dielectric subspace:

$$W^{\text{sym}}(\mathbf{q}, \omega) \approx P(\mathbf{q}) W^{\text{sym}}(\mathbf{q}, \omega) P(\mathbf{q}) \\ = U(\mathbf{q}) W_{\text{aux}}^{\text{sym}}(\mathbf{q}, \omega) U^\dagger(\mathbf{q})$$

with the reduced matrix

$$W_{\text{aux}}^{\text{sym}}(\mathbf{q}, \omega) \equiv U^\dagger(\mathbf{q}) W^{\text{sym}}(\mathbf{q}, \omega) U(\mathbf{q}) \\ \in \mathbb{C}^{N_{\text{aux}} \times N_{\text{aux}}}$$

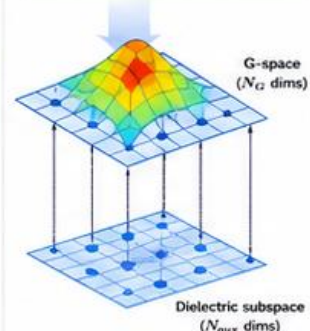


- This is the only approximation in Step 1
- Local-field cost reduced: $N_G \rightarrow N_{\text{aux}}$
- Full $(\mathbf{q}, \omega)$ dependence of screening is retained within the chosen subspace

4c. Reconstruction in G-space

The compressed matrix can be lifted back to G-space if needed:

$$W^{\text{sym}}(\mathbf{q}, \omega) \approx U(\mathbf{q}) W_{\text{aux}}^{\text{sym}}(\mathbf{q}, \omega) U^\dagger(\mathbf{q})$$



- All quantities remain on the real-frequency mesh $\{\omega_i\}$
- No approximation to frequency dependence is introduced in Step 1

5. Convergence diagnostic

Projection error (Frobenius norm):

$$\epsilon_{\text{proj}}(\mathbf{q}, \omega) = \frac{\|W^{\text{sym}}(\mathbf{q}, \omega) - P(\mathbf{q}) W^{\text{sym}}(\mathbf{q}, \omega) P(\mathbf{q})\|_F}{\|W^{\text{sym}}(\mathbf{q}, \omega)\|_F}$$

↓ decreases systematically as N_{aux} is increased.

6. Output for Step 2

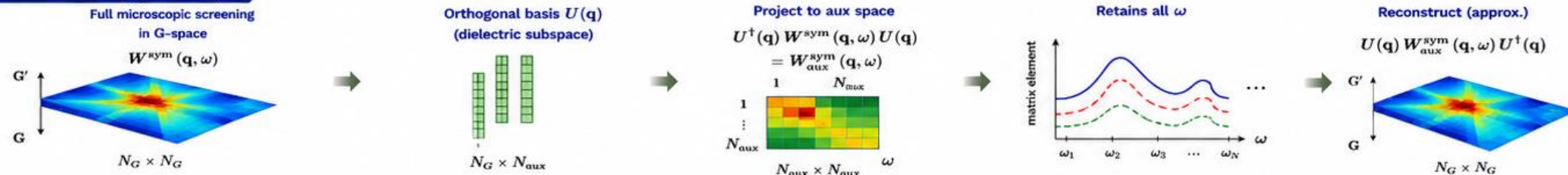
Step 1 outputs the projected symmetrized screened interaction in the aux space on the real-frequency mesh:

$$W_{\text{aux}}^{\text{sym}}(\mathbf{q}, \omega_i) \in \mathbb{C}^{N_{\text{aux}} \times N_{\text{aux}}}$$

for all $\mathbf{q}$ and sampled ω_i .

This is the input for constructing the multi-pole model in Step 2.

Visual summary (for each $\mathbf{q}$)



Key takeaway

Step 1 symmetrizes the microscopic screened interaction and compresses only the local-field (G-space) dimension from N_G to $N_{\text{aux}} \ll N_G$ using an orthogonal dielectric subspace. The full $(\mathbf{q}, \omega)$ dependence of screening is retained within this subspace. The resulting matrices $W_{\text{aux}}^{\text{sym}}(\mathbf{q}, \omega)$ are the direct input for Step 2 (multi-pole compression of the frequency dependence).



Reduce local-field dimension:
 $N_G \rightarrow N_{\text{aux}}$



Full $(\mathbf{q}, \omega)$ dependence preserved



Controlled by projection error
 $\epsilon_{\text{proj}}(\mathbf{q}, \omega)$



Only approximation in Step 1: truncation of the local-field space

STEP 2

Full-frequency screening and scalable multi-pole compression

Compute the symmetrized screened interaction and fit its frequency dependence in the aux space without changing the local-field rank

What we do

We treat the explicit frequency dependence of microscopic screening after the local-field reduction of Step 1.

Symmetrized screened interaction (full G-space)

$$W^{sym}(\mathbf{q}, \omega) \in \mathbb{C}^{N_G \times N_G}$$

Computed from first principles on a real-frequency mesh (e.g., within RPA using contour deformation / numerical integration).

Projection to dielectric (aux) subspace

$$W_{aux}^{sym}(\mathbf{q}, \omega) = U^\dagger(\mathbf{q}) W^{sym}(\mathbf{q}, \omega) U(\mathbf{q}) \in \mathbb{C}^{N_{aux} \times N_{aux}}$$

All quantities in this Step act in the aux space of dimension $N_{aux} \ll N_G$ (fixed by Step 1).

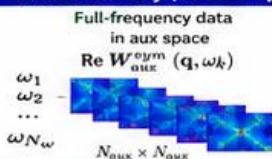
Goal of this step

For each $\mathbf{q}$, fit $W_{aux}^{sym}(\mathbf{q}, \omega)$ in the target frequency window to a causal multi-pole model in the same aux space using a small number of pole pairs N_p . N_p is a frequency-resolution knob (independent of N_{aux}).

Inputs (from Step 1)

- Dielectric subspace basis $U(\mathbf{q}) \in \mathbb{C}^{N_G \times N_{aux}}$
- Fixed subspace dimension $N_{aux} \ll N_G$
- Real-frequency mesh $\{\omega_i\}$ with small $\eta > 0$
- Momentum transfer $\mathbf{q}$

Visual summary (for each $\mathbf{q}$)

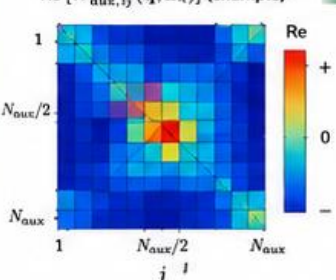


1 Compute data

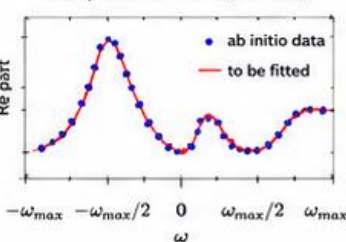
Compute $W_{aux}^{sym}(\mathbf{q}, \omega)$ on a real-frequency grid in the exciton-relevant window.

Matrix dimension: $N_{aux} \times N_{aux}$

Re $[W_{aux,ij}^{sym}(\mathbf{q}, \omega_k)]$ (example)



Example element vs. ω (real axis)



$\eta > 0$ ensures the causal prescription (retarded)

Build the multi-pole model in aux space (for each $\mathbf{q}$)

2 Multi-pole fit (causal)

Fit $W_{aux}^{sym}(\mathbf{q}, \omega)$ to a causal pole expansion in the same aux space.

$$W_{aux}^{sym}(\mathbf{q}, \omega) \approx W_{aux}^{(\infty)}(\mathbf{q}) + \sum_{a=1}^{N_p} S_a^{aux}(\mathbf{q}) f_a(\mathbf{q}; \omega)$$

with

$$f_a(\mathbf{q}; \omega) = \frac{1}{\omega - \omega_a(\mathbf{q}) + i\eta} - \frac{1}{\omega + \omega_a(\mathbf{q}) + i\eta}$$

- $\omega_a(\mathbf{q}) > 0$: pole frequencies
- $\eta \rightarrow 0^+$: causal (retarded) prescription (small numerical broadening in practice)
- N_p : number of pole pairs (accuracy knob)

Goodness of fit (example)

Relative Frobenius error over window:

$$\epsilon_{fit}(\mathbf{q}) = \left(\frac{\sum_k \|W_{aux}^{sym}(\mathbf{q}, \omega_k) - W_{aux}^{model}(\mathbf{q}, \omega_k)\|_F^2}{\sum_k \|W_{aux}^{sym}(\mathbf{q}, \omega_k)\|_F^2} \right)^{1/2}$$

3 Residues and factorization

The residues are Hermitian PSD matrices in aux space.

$$S_a^{aux}(\mathbf{q}) = [S_a^{aux}(\mathbf{q})]^\dagger \geq 0 \in \mathbb{C}^{N_{aux} \times N_{aux}}$$

Factorize each residue (low-rank):

$$S_a^{aux}(\mathbf{q}) = B_a(\mathbf{q}) B_a^\dagger(\mathbf{q})$$

with $B_a(\mathbf{q}) \in \mathbb{C}^{N_{aux} \times r_a}$, $r_a \leq N_{aux}$ (often $r_a \ll N_{aux}$)

$$S_a^{aux}(\mathbf{q}) \quad B_a(\mathbf{q}) \quad B_a^\dagger(\mathbf{q})$$

$$N_{aux} \times N_{aux} \quad N_{aux} \times r_a \quad r_a \times N_{aux}$$

Factorization is always possible for PSD matrices (e.g., eigen-decomposition with thresholding). Key enabler for Step 4.

4 Resulting model

Scalable, causal, compact representation in aux space.

$$W_{aux}^{sym}(\mathbf{q}, \omega) \approx W_{aux}^{(\infty)}(\mathbf{q}) + \sum_{a=1}^{N_p} B_a(\mathbf{q}) B_a^\dagger(\mathbf{q}) \times f_a(\mathbf{q}; \omega)$$

All matrices act in $\mathbb{C}^{N_{aux} \times N_{aux}}$.

Only the frequency dependence is compressed; the local-field rank N_{aux} is unchanged.

Key properties (in aux space)

- ✓ Hermitian in the lossless real-frequency limit (for real ω with $\eta \rightarrow 0^+$): $W_{aux}^{sym}(\mathbf{q}, \omega) = [W_{aux}^{sym}(\mathbf{q}, \omega)]^\dagger$
- ✓ Causal (retarded) structure from pole pairs at $\pm\omega_a(\mathbf{q})$ with same residue.
- ✓ Residues $S_a^{aux}(\mathbf{q}) \geq 0$ (physical screening).
- ✓ Full $(\mathbf{q}, \omega)$ dependence retained within the model (accuracy controlled by N_p).
- ✓ Only approximation: fit to the data in the chosen frequency window.

Convergence diagnostic

Projection error (Frobenius norm)

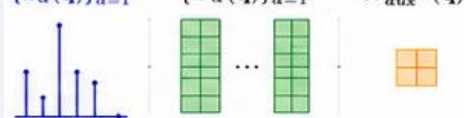
$$\epsilon_{proj}(\mathbf{q}, \omega) = \frac{\|W_{aux}^{sym} - W_{aux}^{model}\|_F}{\|W_{aux}^{sym}\|_F}$$

↓ decreases as N_p increases

Output for Step 2 (per $\mathbf{q}$)

Inputs for Step 3 (exciton-screening couplings):

Pole frequencies $\{\omega_a(\mathbf{q})\}_{a=1}^{N_p}$ Residue factors $\{B_a(\mathbf{q})\}_{a=1}^{N_p}$ Constant term $W_{aux}^{(\infty)}(\mathbf{q})$



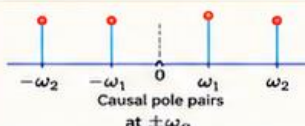
with $S_a^{aux}(\mathbf{q}) = B_a(\mathbf{q}) B_a^\dagger(\mathbf{q}) \geq 0$

Key takeaway

Step 2 represents the full-frequency symmetrized screened interaction in the dielectric (aux) subspace by a compact, causal multi-pole model. The local-field rank N_{aux} remains fixed from Step 1, while the number of pole pairs N_p controls the accuracy of the frequency dependence. The low-rank factorization of the PSD residues is the key ingredient that enables the frequency-free linearization in Step 4.



Real-frequency first-principles data



Hermitian PSD residues



Accuracy controlled by N_p (independent of N_{aux})



Enables efficient subsequent linearization

STEP 3

Exciton-screening couplings and rational dynamical kernel

Couple the multi-pole screening model (from Step 2) to the electron-hole transition space and build the rational kernel

Input from Step 2

Reduced symmetrized screened interaction in the dielectric subspace (for each $\mathbf{q}$)

$$W_{\text{aux}}^{\text{sym}}(\mathbf{q}, \omega) = W_{\text{aux}}^{(\infty)}(\mathbf{q}) + \sum_{\alpha=1}^{N_p} S_{\alpha}^{\text{aux}}(\mathbf{q}) \left[\frac{1}{\omega - \omega_{\alpha}(\mathbf{q}) + i\eta} - \frac{1}{\omega + \omega_{\alpha}(\mathbf{q}) + i\eta} \right]$$

$W_{\text{aux}}^{\text{sym}}(\mathbf{q}, \omega)$: reduced, symmetrized screened interaction

$W_{\text{aux}}^{(\infty)}(\mathbf{q})$: high-frequency (static) limit

$S_{\alpha}^{\text{aux}}(\mathbf{q}) = B_{\alpha}(\mathbf{q}) B_{\alpha}^{\dagger}(\mathbf{q}) \geq 0$
(residue, rank r_{α})

$\omega_{\alpha}(\mathbf{q}) > 0$: pole frequency

N_p : number of pole pairs (kept small)

$\eta > 0$: infinitesimal ensuring causality

1. Transition densities and projection

1 Transition-density matrix (G-resolved)

$$M(\mathbf{q}) \in \mathbb{C}^{N_G \times N_{eh}(\mathbf{q})}, \quad [M(\mathbf{q})]_{G,\nu} = M_{\nu}^G(\mathbf{q})$$

columns $\nu \equiv (v, c, \mathbf{k})$

2 Symmetrized (Coulomb-weighted) densities

$$\tilde{M}_{\nu}^G(\mathbf{q}) = \sqrt{v_G(\mathbf{q})} M_{\nu}^G(\mathbf{q})$$

Collect into $\tilde{M}(\mathbf{q}) \in \mathbb{C}^{N_G \times N_{eh}(\mathbf{q})}$

3 Projection to dielectric subspace

$$\tilde{M}_{\text{aux}}(\mathbf{q}) = U^{\dagger}(\mathbf{q}) \tilde{M}(\mathbf{q})$$

$\in \mathbb{C}^{N_{\text{aux}} \times N_{eh}(\mathbf{q})}$

N_G : number of reciprocal vectors

N_{aux} : dielectric-subspace dimension (Step 1)

$N_{eh}(\mathbf{q})$: number of e-h transitions coupled by $\mathbf{q}$

2. Projected direct kernel in aux space

The projected direct kernel is

$$K^d(\Omega) \approx - \sum_{\mathbf{q}} \tilde{M}_{\text{aux}}^{\dagger}(\mathbf{q}) W_{\text{aux}}^{\text{sym}}(\mathbf{q}, \Omega) \tilde{M}_{\text{aux}}(\mathbf{q})$$

Insert the pole expansion (from Step 2):

$$K^d(\Omega) = K_{\text{stat}}^d + \text{dynamical part}$$

Static (frequency-independent) contribution

$$K_{\text{stat}}^d = - \sum_{\mathbf{q}} \tilde{M}_{\text{aux}}^{\dagger}(\mathbf{q}) W_{\text{aux}}^{(\infty)}(\mathbf{q}) \tilde{M}_{\text{aux}}(\mathbf{q})$$

Define pole couplings in transition space

$$V_{\alpha}(\mathbf{q}) \equiv B_{\alpha}^{\dagger}(\mathbf{q}) \tilde{M}_{\text{aux}}(\mathbf{q})$$

$\in \mathbb{C}^{r_{\alpha} \times N_{eh}(\mathbf{q})}$

with $B_{\alpha}(\mathbf{q}) \in \mathbb{C}^{N_{\text{aux}} \times r_{\alpha}}$

$$(S_{\alpha}^{\text{aux}}(\mathbf{q}) = B_{\alpha}(\mathbf{q}) B_{\alpha}^{\dagger}(\mathbf{q}))$$

$$r_{\alpha} \begin{matrix} N_{\text{aux}} \\ B_{\alpha}(\mathbf{q}) \\ N_{\text{aux}} \end{matrix} \times \begin{matrix} N_{\text{aux}} \\ \tilde{M}_{\text{aux}}(\mathbf{q}) \\ N_{eh}(\mathbf{q}) \end{matrix} = \begin{matrix} r_{\alpha} \\ V_{\alpha}(\mathbf{q}) \\ N_{eh}(\mathbf{q}) \end{matrix}$$

3. Rational dynamical kernel

The dynamical direct kernel becomes

$$K^d(\Omega) = K_{\text{stat}}^d + \sum_{\mathbf{q}} \sum_{\alpha=1}^{N_p} V_{\alpha}^{\dagger}(\mathbf{q}) \Pi_{\alpha}(\mathbf{q}; \Omega) V_{\alpha}(\mathbf{q})$$

static part
(freq.-indep.)

frequency-dependent
(resolvents)

couplings
(freq.-indep.)

All frequency dependence is in the scalar resolvent

$$\Pi_{\alpha}(\mathbf{q}; \Omega) \equiv -f_{\alpha}(\mathbf{q}; \Omega) = - \left[\frac{1}{\Omega - \omega_{\alpha}(\mathbf{q}) + i\eta} - \frac{1}{\Omega + \omega_{\alpha}(\mathbf{q}) + i\eta} \right]$$

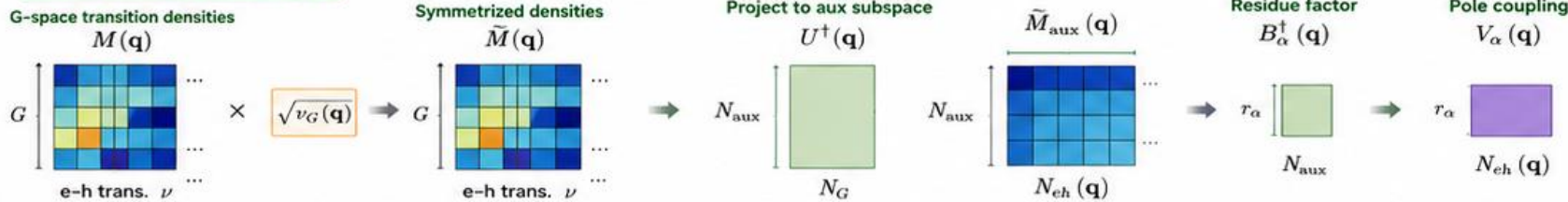
$\Rightarrow$ multiplies the identity in α -subspace of size r_{α}

Key message of Step 3

Once $U(\mathbf{q})$ (Step 1) and $\{\omega_{\alpha}(\mathbf{q}), S_{\alpha}^{\text{aux}}(\mathbf{q})\}$ (Step 2) are fixed,

- all band-structure & local-field information enters only through the frequency-independent couplings $V_{\alpha}(\mathbf{q})$.
- The full dynamical dependence is isolated in the simple rational factors $\Pi_{\alpha}(\mathbf{q}; \Omega)$.

Visual summary (for each $\mathbf{q}$)



Rational kernel assembly

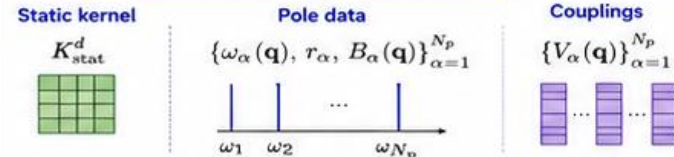
$$K_{\text{stat}}^d + \sum_{\mathbf{q}} \sum_{\alpha} V_{\alpha}^{\dagger} \Pi_{\alpha}(\mathbf{q}; \Omega) V_{\alpha}$$

static (freq.-indep.) resolvent (freq.-dep.) couplings (freq.-indep.)

Checks (for each $\mathbf{q}$)

- $S_{\alpha}^{\text{aux}}(\mathbf{q}) \geq 0$ (Hermitian PSD)
- $W_{\text{aux}}^{\text{sym}}(\mathbf{q}, \omega) = W_{\text{aux}}^{\text{sym}}(\mathbf{q}, -\omega)^{\dagger}$ (Hermitian in lossless limit)
- Poles in pairs at $\pm \omega_{\alpha}(\mathbf{q})$ (causality)
- N_p small, errors below threshold (Step 2 validation)

Output of Step 3 (for each $\mathbf{q}$)



Inputs for Step 4 (frequency-free auxiliary linearization)



Key takeaway

Step 3 couples the multi-pole screening model to the electron-hole transition space and yields a frequency-independent static kernel plus frequency-independent couplings $V_{\alpha}(\mathbf{q})$, while all explicit frequency dependence is contained in simple scalar rational factors:

$$K^d(\Omega) = K_{\text{stat}}^d + \sum_{\mathbf{q}} \sum_{\alpha} V_{\alpha}^{\dagger}(\mathbf{q}) \Pi_{\alpha}(\mathbf{q}; \Omega) V_{\alpha}(\mathbf{q})$$

This is the direct input for the frequency-free auxiliary linearization in Step 4.

STEP 4

Frequency-free auxiliary linearization (single eigenproblem)

Recast the dynamical BSE as a single generalized arrowhead eigenproblem in an enlarged space (no frequency scan)

From the rational BSE (Step 3 output)

After Steps 1–3, the dynamical BSE reads

$$(H_{\text{stat}} - \Omega I) A + \sum_{\mathbf{q}} \sum_{\alpha=1}^{N_p} V_{\alpha}^{\dagger}(\mathbf{q}) \Pi_{\alpha}(\mathbf{q}; \Omega) V_{\alpha}(\mathbf{q}) A = 0$$

with

- $H_{\text{stat}} = D + K^x + K_{\text{stat}}^d \in \mathbb{C}^{N_{eh} \times N_{eh}}$ (static exciton Hamiltonian)
- $A \in \mathbb{C}^{N_{eh}}$ (excitonic amplitude vector)
- $V_{\alpha}(\mathbf{q}) \in \mathbb{C}^{r_{\alpha} \times N_{eh}}$ (pole couplings from Step 3)
- $\Pi_{\alpha}(\mathbf{q}; \Omega) = - \left[\frac{1}{\Omega - \omega_{\alpha}(\mathbf{q}) + i\eta} - \frac{1}{\Omega + \omega_{\alpha}(\mathbf{q}) + i\eta} \right]$ (causal limit: $\eta \rightarrow 0^+$)

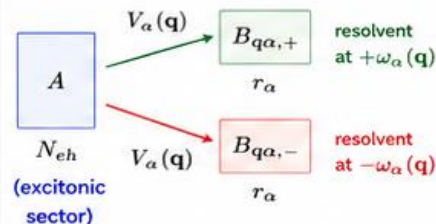
Goal of this step

Eliminate any explicit frequency scan by an exact linearization using auxiliary variables associated with the poles at $\pm \omega_{\alpha}(\mathbf{q})$. This yields a single generalized eigenproblem independent of Ω .

Construct the auxiliary linearization

1 Introduce auxiliary variables

For each $(\mathbf{q}, \alpha)$, introduce two auxiliary vectors of size r_{α} : $B_{q\alpha,+}$ and $B_{q\alpha,-}$, associated with $\pm \omega_{\alpha}(\mathbf{q})$.



Collect all unknowns in the augmented vector

$$\mathbf{X} = (A, \{B_{q\alpha,+}\}, \{B_{q\alpha,-}\})^T$$



2 Define coupling and pole matrices

- Stack pole-coupling blocks vertically

$$V = \begin{pmatrix} V_1(\mathbf{q}_1) \\ \vdots \\ V_{\alpha}(\mathbf{q}) \\ \vdots \end{pmatrix} \in \mathbb{C}^{(\sum_{\mathbf{q},\alpha} r_{\alpha}) \times N_{eh}}$$

- Pole-frequency matrix (positive poles)

$$\Omega_{+} = \text{diag}(\omega_{\alpha}(\mathbf{q}) I_{r_{\alpha}})_{\mathbf{q},\alpha} \in \mathbb{C}^{(\sum_{\mathbf{q},\alpha} r_{\alpha}) \times (\sum_{\mathbf{q},\alpha} r_{\alpha})}$$

- r_{α} : rank of residue factor B_{α}
- N_{eh} : number of e-h transitions
- N_p : number of pole pairs (kept small)

3 Generalized arrowhead eigenproblem

The rational BSE is exactly equivalent to

$$\mathcal{H}_{\text{aux}} \mathbf{X} = \Omega \mathcal{S} \mathbf{X}$$

with

$$\mathcal{H}_{\text{aux}} = \begin{pmatrix} H_{\text{stat}} & -V^{\dagger} & +V^{\dagger} \\ V & \Omega_{+} & 0 \\ -V & 0 & \Omega_{+} \end{pmatrix} \quad \mathcal{S} = \text{diag}(I_{N_{eh}}, I, -I)$$

Arrowhead structure

	excitonic sector	auxiliary (+) sector	auxiliary (-) sector
excitonic (N_{eh})	H_{stat}	$-V^{\dagger}$	$+V^{\dagger}$
(+)	V	Ω_{+}	0
(-)	$-V$	0	Ω_{+}

Dimension of the problem

The generalized eigenproblem has size

$$\dim(\mathcal{H}_{\text{aux}}) = N_{eh} + 2 \sum_{\mathbf{q}} \sum_{\alpha=1}^{N_p} r_{\alpha}$$

If all residues are full rank $r_{\alpha} = N_{aux}$, this reduces to

$$N_{eh} + 2N_p N_{aux}$$

- Well suited to iterative eigensolvers targeting a few low-lying excitons.

- Converge N_{aux} using $\epsilon_{\text{proj}}(\mathbf{q}, \omega)$ (Eq. $(W_{\text{error}}^{\text{proj}})$).

- Converge N_p using the fidelity of the pole fit of $W_{\text{aux}}^{\text{sym}}(\mathbf{q}, \omega)$.

Equivalence: eliminating auxiliary variables recovers the rational BSE

From 2nd block row:

$$(\Omega I - \Omega_{+}) B_{+} = V A \\ \Rightarrow B_{+} = (\Omega - \Omega_{+})^{-1} V A$$

From 3rd block row:

$$(\Omega I + \Omega_{+}) B_{-} = -V A \\ \Rightarrow B_{-} = (\Omega + \Omega_{+})^{-1} V A$$

$$A = \begin{pmatrix} B_{+} \\ B_{-} \end{pmatrix} = \begin{pmatrix} (\Omega - \Omega_{+})^{-1} V A \\ (\Omega + \Omega_{+})^{-1} V A \end{pmatrix} = \begin{pmatrix} (\Omega - \omega_1)^{-1} I_{r_1} & (\Omega - \omega_2)^{-1} I_{r_2} & \dots & (\Omega - \omega_M)^{-1} I_{r_M} \\ (\Omega + \omega_1)^{-1} I_{r_1} & (\Omega + \omega_2)^{-1} I_{r_2} & \dots & (\Omega + \omega_M)^{-1} I_{r_M} \end{pmatrix} V A$$

Insert back into the first block row:

$$(H_{\text{stat}} - \Omega I) A + V^{\dagger} B_{+} - V^{\dagger} B_{-} = 0 \\ \Rightarrow (H_{\text{stat}} - \Omega I) A + \sum_{\mathbf{q}} \sum_{\alpha=1}^{N_p} V_{\alpha}^{\dagger}(\mathbf{q}) \Pi_{\alpha}(\mathbf{q}; \Omega) V_{\alpha}(\mathbf{q}) A = 0$$

(exactly the rational BSE)

Thus, the linearization is exact

The only approximations are

- Dielectric-subspace truncation (Step 2, Eq. $(W_{\text{project}}^{\text{sym}})$).
- Multi-pole compression of $W_{\text{aux}}^{\text{sym}}(\mathbf{q}, \omega)$ (Step 3, Eq. $(W_{\text{multipole}}^{\text{aux}})$)

Key takeaway

Step 4 converts the frequency-dependent (rational) BSE into a single, frequency-free generalized arrowhead eigenproblem in an enlarged space. This avoids any iterative search over Ω and enables efficient computation of excitons with controlled, systematically improvable approximations.



No frequency scan:
solve once



Structured
arrowhead
matrix



Targets low-lying
excitons efficiently



Exact reformulation
(with controlled
approximations)



DYNAMO reads Abinit data and computes BSE results



Inputs from ABINIT



*_SCR.nc

Screening on real frequency (optdriver = 3)

- q-points $\mathbf{q}$
- frequency grid ω
- inverse dielectric matrix $\epsilon_{GG'}^{-1}(\mathbf{q}, \omega)$ (symmetrized)
- G-vectors $\mathbf{G}$



*_WFK

(binary wavefunctions)

- Bloch states ψ_{nk}
- FFT mesh information



H_resonant.nc

Static resonant BSE Hamiltonian (TDA block)
 H_{stat} in transition space

Additional files (optional)



*_EIG.nc ϵ_{nk} (eigenvalues)



*_DEN.nc ρ_{GS} (ground-state density)

DYNAMO workflow

1 Export $W^{sym}(\mathbf{q}, \omega)$ from SCR



W_DATA

For each $\mathbf{q}$, build the symmetrized screened interaction in G-space on the real-frequency grid:

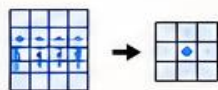
$$W_{GG'}^{sym}(\mathbf{q}, \omega) = \sqrt{v_G(\mathbf{q})} \epsilon_{GG'}^{-1}(\mathbf{q}, \omega) \sqrt{v_{G'}(\mathbf{q})}$$

$$v_G(\mathbf{q}) = \frac{4\pi}{|\mathbf{q} + \mathbf{G}|^2}$$

Output per $\mathbf{q}$:

$$W_{GG'}^{sym}(\mathbf{q}, \omega), \omega_j, \mathbf{G}, \mathbf{q}$$

2 Build $U(\mathbf{q})$ and $W_{aux}(\mathbf{q}, \omega)$



U_WAUX

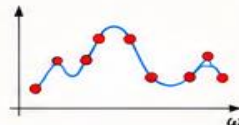
Choose an orthonormal dielectric subspace $U(\mathbf{q}) \in \mathbb{C}^{N_G \times N_{aux}}$ ($U^\dagger U = I$) and project:

$$W_{aux}^{sym}(\mathbf{q}, \omega) = U^\dagger(\mathbf{q}) W^{sym}(\mathbf{q}, \omega) U(\mathbf{q})$$

Outputs per $\mathbf{q}$:

$$U(\mathbf{q}), W_{aux}(\mathbf{q}, \omega)$$

3 Fit poles $S_\alpha(\mathbf{q})$, $\omega_\alpha(\mathbf{q})$



POLES

Fit $W_{aux}^{sym}(\mathbf{q}, \omega)$ in the exciton window to a multi-pole model:

$$W_{vck}^{sym}(\mathbf{q}, \omega) \approx W_{aux}^{(\infty)}(\mathbf{q}) + \sum_{\alpha=1}^{N_p} S_\alpha^{aux}(\mathbf{q}) \left[\frac{1}{\omega - \omega_\alpha + i\eta} - \frac{1}{\omega + \omega_\alpha + i\eta} \right]$$

Outputs per $\mathbf{q}$:

$$\omega_\alpha(\mathbf{q}), S_\alpha(\mathbf{q}) \text{ (or factors } B_\alpha)$$

4 Build $M_{aux}(\mathbf{q})$ from WFK



MAUX

Build transition densities in G-space from *_WFK, then project with $U(\mathbf{q})$:

$$\tilde{M}_{vck}^G(\mathbf{q}) = \sqrt{v_G(\mathbf{q})} M_{vck}^G(\mathbf{q})$$

$$\tilde{M}_{aux}(\mathbf{q}) = U^\dagger(\mathbf{q}) \tilde{M}(\mathbf{q})$$

Output per $\mathbf{q}$:

$$M_{aux}(\mathbf{q}) \in \mathbb{C}^{N_{aux} \times N_{eh}(\mathbf{q})}$$

5 Build $V_\alpha(\mathbf{q})$ (pole couplings)



VALPHA

Using $S_\alpha = B_\alpha B_\alpha^\dagger$, compute pole couplings:

$$V_\alpha(\mathbf{q}) = B_\alpha^\dagger(\mathbf{q}) \tilde{M}_{aux}(\mathbf{q})$$

Output per $\mathbf{q}$:

$$\omega_\alpha(\mathbf{q}), r_\alpha, V_\alpha(\mathbf{q}) \text{ (padded to constant rank)}$$

6 Assemble and solve (auxiliary problem)



AUX_SOLVE

Solve the frequency-free dynamical BSE by linearizing to an auxiliary generalized eigenproblem:

$$(H_{stat} - \Omega I)A + \sum_{\mathbf{q}, \alpha} V_\alpha^\dagger(\mathbf{q}) \Pi_\alpha(\mathbf{q}; \Omega) V_\alpha(\mathbf{q}) A = 0$$

Output:

Dynamical eigenvalues Ω and excitonic properties

Results computed by DYNAMO



Dynamical exciton energies Ω_S



Exciton wavefunctions A_{vck}^Ω



Oscillator strengths f_s (optical spectra)



Absorption spectrum $\alpha(\omega)$, $\text{Im } \epsilon_M(\omega)$



Momentum / energy maps $A(\mathbf{k}, \omega)$



Residuals, exciton weights, nearest static eigenvalues, dynamical shifts
 $\Delta\Omega = \text{Re } \Omega - \lambda_{nearest}$

Typical Abinit inputs required

- ✓ Pseudopotentials (UPF)
- ✓ SCF input for ground state
- ✓ NSCF input for unocc. states
- ✓ k-mesh, energy cutoff, smearing
- ✓ Dielectric (RPA) settings (optdriver = 3, real-frequency grid)



Data flow

*_SCR.nc
*_WFK
+ H_resonant.nc
(+ optional)
ABINIT outputs



- W_DATA
- U_WAUX
- POLES
- MAUX
- VALPHA
- AUX_SOLVE

DYNAMO is a many-body post-processing code

- ✓ Reads ABINIT outputs (*_SCR.nc, *_WFK, H_resonant.nc)
- ✓ Uses the same pseudopotentials & conventions
- ✓ No re-run of DFT is needed
- ✓ Multi-pole compression of screened interaction
- ✓ Frequency-free solution of the BSE
- ✓ High performance & parallel (PETSc/SLEPc + MUMPS)



From ABINIT you take $\epsilon^{-1}(\mathbf{q}, \omega)$, G-vectors and wavefunctions; you build $W^{sym}(\mathbf{q}, \omega)$, compress local fields with $U(\mathbf{q})$ to get $W_{aux}(\mathbf{q}, \omega)$, fit to a small set of poles, compute transition couplings $V_\alpha(\mathbf{q})$, and finally solve the frequency-free auxiliary eigenproblem to obtain dynamical exciton energies and shifts.



THANK YOU

FOR YOUR ATTENTION